

Salient Characteristics: Backup Fire Suppression System

1. Define the Need

The purpose of this acquisition is to procure a reliable, automated, and self-contained backup fire suppression capability for a high-value aircraft hangar at Fort Cavazos, which houses approximately \$280 million in military assets. The primary fire suppression system does not adequately protect certain "dead space" areas within the hangar. This procurement addresses the critical need to mitigate the risk of catastrophic asset loss in the event of a fire where the primary system fails or is insufficient.

2. Salient Characteristics Table

Category	Characteristic	Specification	Mandatory/ Desirable	Rationale
Performance	Suppression Area	The system shall be capable of effectively protecting a minimum area of 7,800 square feet per unit.	Mandatory	Required to cover the designated "dead space" zones within the hangar and provide overlapping coverage.
Performance	Foam Capacity	Each unit shall have a minimum foam capacity of 240 gallons.	Mandatory	Ensures a sufficient volume of suppression agent is available to extinguish a potential aircraft fuel fire.
Functional	Operation Mode	The system shall be completely self-contained, unmanned, and capable of automatic deployment upon fire detection.	Mandatory	Critical for providing immediate backup coverage in the event of primary system failure when personnel may not be present.
Functional	Discharge Mechanism	Each unit shall be equipped with one (1) or two (2) oscillating turrets.	Mandatory	Ensures comprehensive and automatic distribution of the fire suppression agent across the coverage area without human intervention.
Functional	Suppression Agent Type	The system shall utilize a cold, dense, high-energy foam capable of creating a superior vapor seal on potential fuel spills.	Mandatory	This specific foam type is required to effectively combat and secure hazardous aviation fuel fires.
Physical	Power Source	The system shall be battery-powered.	Mandatory	Ensures operational capability independent of facility power, which may be lost or compromised during a fire event.
Logistics	Quantity	The procurement shall be for a quantity of three (3) units.	Mandatory	Necessary to provide the required overlapping coverage for all identified unprotected areas within the hangar.
Interface	System Integration	The system should be easily integrated with existing facility fire alarm and detection systems.	Desirable	Allows for seamless activation and centralized monitoring alongside the hangar's primary safety systems.
Logistics	Maintenance Support	A comprehensive maintenance and servicing agreement should be an available option.	Desirable	To ensure the units remain serviceable, certified, and ready for immediate deployment throughout their lifecycle.

Physical	Environmental	The system should utilize environmentally friendly fire suppression agents where possible.	Desirable	To align with modern DoD environmental standards and reduce the impact of hazardous materials.
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